What is Bootstrap?

- Bootstrap is a free front-end framework for faster and easier web development
- Bootstrap includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional JavaScript plugins
- Bootstrap also gives you the ability to easily create responsive designs

What is Responsive Web Design?

Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops.

Why Use Bootstrap?

Advantages of Bootstrap:

- **Easy to use:** Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- **Responsive features:** Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- **Mobile-first approach:** In Bootstrap 3, mobile-first styles are part of the core framework
- **Browser compatibility:** Bootstrap is compatible with all modern browsers (Chrome, Firefox, Internet Explorer, Edge, Safari, and Opera)

Bootstrap 3 is designed to be responsive to mobile devices. Mobile-first styles are part of the core framework.

To ensure proper rendering and touch zooming, add the following `<meta>` tag inside the `<head>` element:

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

The `width=device-width` part sets the width of the page to follow the screen-width of the device (which will vary depending on the device).

The `initial-scale=1` part sets the initial zoom level when the page is first loaded by the browser.

Containers

Bootstrap also requires a containing element to wrap site contents.

There are two container classes to choose from:

1. The `.container` class provides a responsive **fixed width container. Gives margin to left and right side.**
2. The `.container-fluid` class provides a **full width container**, spanning the entire width of the viewport.

Bootstrap Grid System

Bootstrap's grid system allows up to 12 columns across the page. If you do not want to use all 12 columns individually, you can group the columns together to create wider columns. Bootstrap's grid system is responsive, and the columns will re-arrange automatically depending on the screen size.

span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1
span 4				span 4				span 4			
span 4				span 8							
span 6						span 6					

Grid Classes

The Bootstrap grid system has four classes:

- `xs` (for phones - screens less than 768px wide)
- `sm` (for tablets - screens equal to or greater than 768px wide)
- `md` (for small laptops - screens equal to or greater than 992px wide)
- `lg` (for laptops and desktops - screens equal to or greater than 1200px wide)

The classes above can be combined to create more dynamic and flexible layouts.

Basic Structure of a Bootstrap Grid

The following is a basic structure of a Bootstrap grid:

```
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
<div class="row">
  ...
</div>
```

First; create a row (`<div class="row">`). Then, add the desired number of columns (tags with appropriate `.col-*-*`classes). Note that numbers in `.col-*-*` should always add up to 12 for each row.

The following example shows how to get two various-width columns starting at tablets and scaling to large desktops:

Example

```
<div class="row">
  <div class="col-sm-4">.col-sm-4</div>
  <div class="col-sm-8">.col-sm-8</div>
</div>
```

Bootstrap's Default Settings

Bootstrap's global default font-size is 14px, with a line-height of 1.428. This is applied to the `<body>` element and all paragraphs (`<p>`). In addition, all `<p>` elements have a bottom margin that equals half their computed line-height (10px by default).

Bootstrap Basic Table

A basic Bootstrap table has a light padding and only horizontal dividers. The `.table` class adds basic styling to a table:

Example

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Condensed Table

The `.table-condensed` class makes a table more compact by cutting cell padding in half:

Example

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Responsive Tables

The `.table-responsive` class creates a responsive table. The table will then scroll horizontally on small devices (under 768px). When viewing on anything larger than 768px wide, there is no difference:

Example

```
<div class="table-responsive">
  <table class="table">
    ...
  </table>
</div>
```

Responsive Images

Images come in all sizes. So do screens. Responsive images automatically adjust to fit the size of the screen. Create responsive images by adding an `.img-responsive` class to the `<img>` tag. The image will then scale nicely to the parent element. The `.img-responsive` class applies `display: block;` and `max-width: 100%;` and `height: auto;` to the image:

Example

```
<img class="img-responsive" src="img_chania.jpg" alt="Chania">
```

Class	Description
.img-rounded	Adds rounded corners to an image (not available in IE8)
.img-circle	Shapes the image to a circle (not available in IE8)
.img-thumbnail	Shapes the image to a thumbnail
.img-responsive	Makes an image responsive (will scale nicely to the parent element)

Button Styles

Bootstrap provides different styles of buttons:

To achieve the button styles above, Bootstrap has the following classes:

- `.btn`
- `.btn-default`
- `.btn-primary`
- `.btn-success`
- `.btn-info`
- `.btn-warning`
- `.btn-danger`
- `.btn-link`

The following example shows the code for the different button styles:

Example

```
<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-default">Default</button>
```

```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-link">Link</button>
```

The class `.active` makes a button appear pressed, and the class `.disabled` makes a button unclickable:

Example

```
<button type="button" class="btn btn-primary active">Active Primary</button>
<button type="button" class="btn btn-primary disabled">Disabled Primary</button>
```

Forms

Basic example

Individual form controls automatically receive some global styling. All textual `<input>`, `<textarea>`, and `<select>` elements with `.form-control` are set to `width: 100%;` by default. Wrap labels and controls in `.form-group` for optimum spacing.

https://getbootstrap.com/docs/4.0/components/forms/

https://www.w3schools.com/bootstrap/bootstrap_forms.asp

EXAMPLE

Email address

Email

Password

Password

File input

Browse... No file selected.

Example block-level help text here.

☐ Check me out

Submit

Top of Form

```
<form>
  <div class="form-group">
    <label for="exampleInputEmail1">Email address</label>
    <input type="email" class="form-control" id="exampleInputEmail1"
placeholder="Email">
  </div>
  <div class="form-group">
    <label for="exampleInputPassword1">Password</label>
    <input type="password" class="form-control" id="exampleInputPassword1"
placeholder="Password">
  </div>
  <div class="form-group">
    <label for="exampleInputFile">File input</label>
    <input type="file" id="exampleInputFile">
    <p class="help-block">Example block-level help text here.</p>
  </div>
  <div class="checkbox">
    <label>
      <input type="checkbox"> Check me out
    </label>
  </div>
  <button type="submit" class="btn btn-default">Submit</button>
</form>
```